

Midterm Review

COMP-550

Readings: None

question 21 from Fall 2023

a)

Start: S

N: {A, B}

T: {x, y}

$S \rightarrow AS$

$A \rightarrow x$

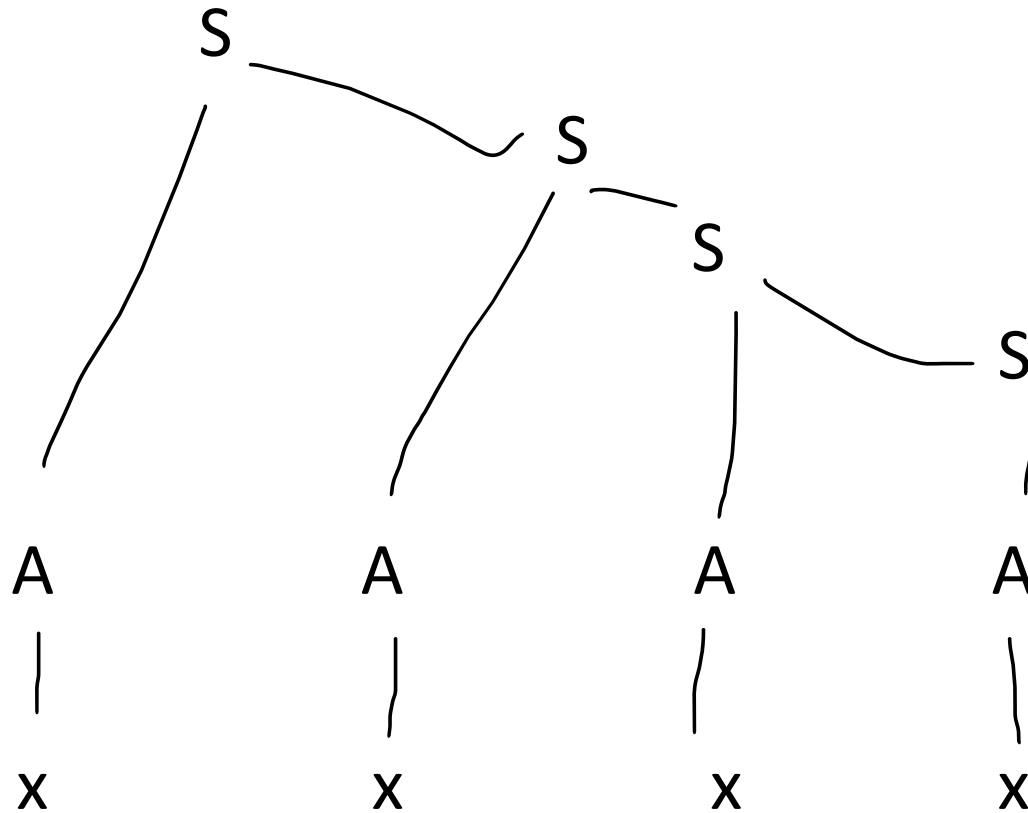
$S \rightarrow A$

$S \rightarrow BS$

$B \rightarrow x$

$S \rightarrow B$

b)



question 25 from Fall 2019 midterm

- a) $\exists x. (Student(x) \wedge \exists y. (Student(y) \wedge x \neq y \wedge Clever(x) \wedge Clever(y)))$
- b) We can't model "most" with FOL because "most" requires knowing how many relevant objects in the world there are and computing a property which involves a proportion of the global set.

2023 #18

a) Vanishing gradient: repeated multiplication of partial derivatives with norm < 1 resulting in a gradient with norm almost equal to zero during backpropagation.

Exploding gradient: repeated multiplication of partial derivatives with norm > 1 resulting in a gradient with norm that is very large during backpropagation.

b) Vanishing: no learning

Exploding: jumping around optimum (overshoot all the time); numerical stability issues

c) Normalizing the gradient; using the LSTM architecture; gradient clipping

2023 #17

a) i) Some teacher teaches difficult subjects

$$\exists x. Teacher(x) \wedge \exists y. Subject(y) \wedge Teaches(x, y) \wedge Difficult(y)$$

ii) Some teacher teaches multiple subjects

$$\exists x. Teacher(x) \wedge \exists y. Subject(y) \wedge Teaches(x, y) \wedge \exists z. Subject(z) \wedge z \neq y \wedge Teaches(x, z)$$

b)

Syntactically different:

something is a difficult subject

*something is a multiple subject

Answer to How many? (semantic? Syntactic?):

Multiple

#Difficult

Very difficult, *very multiple

Multiple: “objective”, quantifiable; Difficult is subjective and vague

#19:

$$\text{Precision(A)} = \text{TP} / (\text{TP} + \text{FP}) = 95 / (95 + 8 + 3)$$

$$\text{Precision(B)} = 40 / (9 + 40 + 3)$$

$$\text{Precision(C)} = 55 / (4 + 4 + 55)$$

$$\text{MacroPrc} = (\text{Prc(A)} + \text{Prc(B)} + \text{Prc(C)}) / 3$$

$$\text{Recall(A)} = \text{TP} / (\text{TP} + \text{FN}) = 95 / (95 + 9 + 4)$$

$$\text{Recall(B)} = 40 / (8 + 40 + 4)$$

$$\text{Recall(C)} = 55 / (3 + 3 + 55)$$

#22:

a)

Generative model: specifies $P(X,Y)$ where X is some observed variable (e.g. words), and Y is some latent variable (e.g. a label or class)

Discriminative model: specifies $P(Y | X)$

b) Generative: HMM, Naïve Bayes

Discriminative: CRF, SVM, Linear Regression

c) Language modelling; Generation by sampling from the joint distribution

2019 # 21

a) $P(\mathbf{X}) = \prod P(x_t | x_{t-1} \dots x_{t-N+1})$ for t from 0 to $|\mathbf{X}|-1$

- Be careful of corner case of $N = 1$

b) Advantage: Captures longer context

Disadvantage: Susceptible to sparsity if dataset is small ; Needs better / more sophisticated smoothing algorithms

#22

$$P_{\text{init}}(C) = (1+1) / (3+4)$$

$$P_{\text{init}}(N) = (0+1) / (3+4)$$

$$P_{\text{init}}(V) = (2+1) / (3+4)$$

$$P_{\text{init}}(J) = (0+1) / (3+4)$$

State transition probabilities

Word emission probabilities (from each state)